

Proposition A.2. *Let $q > n$, let $\Omega \subset \mathbb{R}^n$ be an open bounded connected set, and let $u \in W^{3,q}(\Omega)$ be a function such that for some positive constant k*

$$|\nabla u|(x) \geq k, \quad \forall x \in u^{-1}(0).$$

Assume in addition that $d(u^{-1}(0), \partial\Omega) > 0$. Let $u_j \in C^{2,\alpha}(\Omega)$ be a sequence of functions such that

$$u_j \xrightarrow{W^{2,q}(\Omega)} u.$$

Then for every j large enough $u_j^{-1}(0)$ is a $C^{1,s}$ hypersurface, with $s = 1 - n/q$, and there exist deformations $\Phi_j \in W^{2,q}(\Omega; \mathbb{R}^n)$ such that

(i) Φ_j are orthogonal to $u^{-1}(0)$;

(ii) $u_j^{-1}(0) = (\text{Id} + \Phi_j)(u^{-1}(0))$;

(iii) $\lim_j \|\Phi_j\|_{2,q} = 0$

Proof. • **Step 1:** we first notice that for every $\varepsilon > 0$ there exists t_0 such that if $|t| \leq t_0$ then

$$u^{-1}(t) \subseteq (u^{-1}(0))^\varepsilon,$$

where we use the notation $(K)^t = \{p \in \mathbb{R}^n \mid d(p, K) \leq t\}$, for the *outer parallel* set.

Indeed, let us assume by contradiction that there exist $\varepsilon_0 > 0$ and points $y_k \in u^{-1}(t_k)$ with $\lim_k t_k = 0$ such that

$$\forall k \in \mathbb{N}, \quad d(y_k, u^{-1}(0)) > \varepsilon_0.$$

As Ω is bounded, up to a subsequence, we may assume that y_k converge to some point $\bar{y} \in \bar{\Omega}$. The continuity of u gives $u(\bar{y}) = 0$, so that $\bar{y} \in u^{-1}(0)$. On the other hand,

$$d(\bar{y}, u^{-1}(0)) \geq \varepsilon_0,$$

which is a contradiction.

• **Step 2:** We now want to extend the non-degeneracy property of the gradient of u to the functions u_j on their level sets $u_j^{-1}(0)$. We notice that by uniform convergence and by the previous step, for every $\varepsilon > 0$ there exist $t_0 > 0$ and j_0 such that

$$\forall j \geq j_0, \quad u_j^{-1}(0) \subset u^{-1}((-t_0, t_0)) \subset (u^{-1}(0))^\varepsilon. \quad (\text{A.2})$$

On the other hand, ∇u is uniformly continuous in Ω so there exists $\alpha_0 > 0$ such that

$$\forall (x, z) \in \Omega \text{ s.t. } |z - x| \leq \alpha_0, \quad |\nabla u(z) - \nabla u(x)| \leq \frac{k}{4}.$$

In particular,

$$\forall x \in (u^{-1}(0))^{\alpha_0}, \quad |\nabla u(x)| \geq \frac{k}{2}.$$

Also, ∇u_j uniformly converge to ∇u , so for j large enough

$$|\nabla u_j(x) - \nabla u(x)| \leq \frac{k}{8} \quad \forall x \in \Omega.$$

From the previous estimates, we get $|\nabla u_j|(x) > 0$ for every $x \in u_j^{-1}(0)$ and j large enough, which implies that the sets $u_j^{-1}(0)$ are $C^{1,s}$ hypersurfaces.